CONCEPT NOTE

Student Performance Prediction Using Machine Learning

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1. Introduction

Student performance can be influenced by several factors, including study time, attendance, and previous academic results. This project demonstrates a simple machine-learning workflow that estimates final marks from these inputs. The aim is to learn how data, model training, and predictions work together in an applied AI project.

2. Problem Statement

Students and educators may find it useful to explore how learning-related factors relate to academic results. This project builds an educational prototype that estimates a possible final mark from a small set of inputs. It is not intended to judge students or make real academic decisions.

3. Objectives

• Build a simple regression model using Python and scikit-learn.
• Use study hours, attendance, and previous marks as input features.
• Create an interactive interface using Streamlit.
• Display an estimated mark and basic model evaluation metrics.
• Understand the limitations of small datasets and machine-learning predictions.

4. Proposed Methodology

The project uses a small synthetic dataset created for demonstration. The data is loaded with Pandas, divided into training and testing sets, and used to train a Random Forest Regressor. The model is evaluated using Mean Absolute Error and R^2 on a held-out split. A Streamlit interface allows a user to enter example values and view an estimate.

5. Tools and Technologies

Python, Pandas, scikit-learn, Streamlit, CSV data, and GitHub for source-code hosting.

6. Expected Outcome

A working educational prototype that accepts three inputs and returns an estimated final mark, shows sample data, and presents basic model information. The repository contains source code, dependencies, dataset, and setup instructions.

7. Limitations and Responsible Use

The dataset is synthetic and small, so results do not represent real students and model metrics may be unstable. Predictions are not guarantees and must not be used for grading, admissions, or any consequential decision. Real-world use would require consent, suitable representative data, privacy protection, bias checks, and careful validation.

8. Conclusion

This project provides a beginner-friendly example of an end-to-end machine-learning workflow, from data preparation and model training to an interactive prediction interface. It is designed for learning and demonstration.